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Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: VII **Section:** C

Project Group ID : CSDMP230037

Network Intrusion Detection System with Explainable AI (CyberSleuth)

Abstract:

Network attacks are increasing every year, and most intrusion detection tools only label traffic as normal or malicious without saying why. Analysts are then left to guess the cause and the next step, and students or small organisations often have no tool they can read and trust. This project, CyberSleuth, aims to build a network intrusion detection system that detects attacks and also explains each alert in plain English.

The system is trained on the CICIDS2017 dataset and detects four attack types: DoS, DDoS, Port Scan and Brute Force. After the flow records are cleaned and prepared, Random Forest and XGBoost classifiers are trained and compared, and the better model is used for detection. SHAP is used to show which traffic features pushed a prediction towards an attack. Each alert is given a confidence level and paired with a recommended action. For live testing, one laptop captures network packets while a second laptop simulates attacks, and simple behaviour rules for port-scan and login-attempt rates work alongside the model. Everything is presented on a Streamlit dashboard.

The expected outcome is a working detector that reaches over 99% accuracy on the test data, gives a readable reason for every alert, and runs in a live two-laptop demonstration. The project should help analysts and students understand what happened, how serious it is and what to do next.

This work supports three Sustainable Development Goals. SDG 9 (Industry, Innovation and Infrastructure): it helps protect digital infrastructure from attacks that disrupt services. SDG 16 (Peace, Justice and Strong Institutions): it helps institutions guard their data and respond to cybercrime. SDG 4 (Quality Education): its plain-English explanations help students learn how attacks work.

Keywords: Network Intrusion Detection, Machine Learning, Explainable AI, SHAP, Cybersecurity, Python

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor: Ms. Zoya Tabassum

Signature: _____ **Date**: _____

Coordinator **Head of the Department**